

Assignment – 05: Spacers

This report explores two crucial aspects of FET fabrication: the importance of spacers and the orientation of fins and spacers. Spacers, typically made from insulating materials, play a critical role in defining the channel region of the FET and ensuring proper electrical isolation. The abstract will delve into the functionalities of spacers, including their impact on transistor performance and scaling limitations. Additionally, the orientation of fins and spacers, a crucial factor in device design, will be discussed.

Question 1: Spacer

In this part of assignment, we will explore the deposition technique employed for spacer formation, investigate the importance of conformality, and analyse the effects of varying etch parameters. The goal is to observe the effects of over-etching the spacer layer.

Figure 1 shows reference image for spacer.

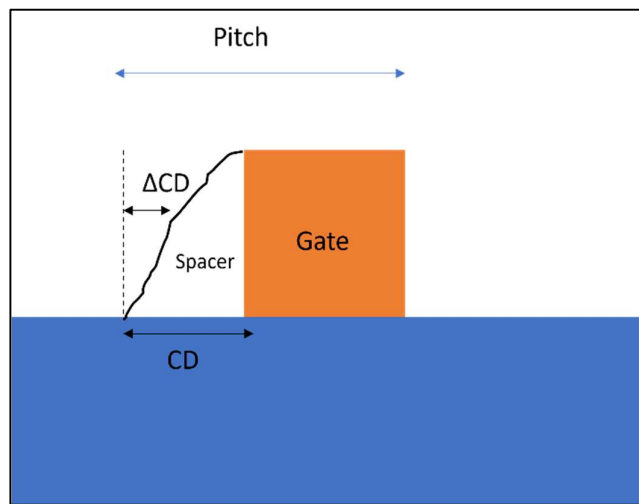


Figure 1 : Reference image for Spacers

The spacer material used usually is Si_3N_4 (silicon nitride) which helps us to properly dope Si for electrical functioning.

Spacer formation done by process flow provided to us is shown by Figure 2 and Figure 3 shows the final structure.

Process Library									
Number	Lock	Action	Name	Material(s)	Dopant(s)	Depth	Mask	Mask Polarity	Wafer / Wafer Side
1		Wafer Setup	Fin formation	/Silicon/Si		100			Silicon1 Top
2.1		Conformal Deposit	Silicon Nitride hard mask for Fin etch	/Nitrides/Si3N4 LPCVD		20			Silicon1 Top
2.2		Planarizing Deposit	resist for SiN hard mask pattern	/Organics/Resist		20			Silicon1 Top
2.3		Expose Material	SiN hard mask pattern	/Organics/Resist		20	Fin_Pattern	Light	Silicon1 Top
2.4		Enhanced Basic Etch	SiN hard mask etch	/Nitrides/Si3N4 LPCVD /Organics/Resist		20			Silicon1 Top
2.5		Remove resist	Remove resist	/Organics/Resist					Silicon1 Top
2.6		Taper Etch	Etch for Fin formation	/Silicon1		60			Silicon1 Top
2.7		Remove Materials	Remove SiN hard mask	/Nitrides/Si3N4 LPCVD					Silicon1 Top
3		Sequence	STI						
3.1		Planarizing Deposit	Planar Deposit	/Oxides/SiO2_HDP		0			Silicon1 Top
3.2		Enhanced Basic Etch	Basic Etch	/Oxides/SiO2_HDP		30			Silicon1 Top
4		Sequence	Source/Drain formation						
4.1		Planarizing Deposit	Resist Deposit for S/D implant	/Organics/Resist		10			Silicon1 Top
4.2		Expose Material	Expose resist for S/D doping	/Organics/Resist		10	S_D_implantation	Dark	Silicon1 Top
4.3		Implant (Advanced Modeling)	Source/Drain Implantation	/Organics/Resist	/NtypeP				Silicon1 Top
4.4		Remove Materials	Remove resist	/Organics/Resist					Silicon1 Top
4.5		Diffusion (Advanced Modeling)	S/D Activation Anneal	/Nitrides /Oxides /Silicon1	/NtypeP				Silicon1 Top
5		Sequence	Gate oxide pattern						
5.1		Conformal Deposit	Gate high k	/Oxides/HfO2		3			Silicon1 Top
5.2		Planarizing Deposit	resist for gate metal pattern	/Organics/Resist		10			Silicon1 Top
5.3		Expose Material	Expose resist	/Organics/Resist		40	Gate	Light	Silicon1 Top
5.4		Enhanced Basic Etch	high k Etch	/Organics/Resist /Oxides/HfO2		3			Silicon1 Top
5.5		Remove Materials	Remove resist	/Organics/Resist					Silicon1 Top
6		Sequence	Gate metal pattern						
6.1		Conformal Deposit	Gate metal deposition	/Metals/Al		5			Silicon1 Top
6.2		Planarizing Deposit	resist for gate metal pattern	/Organics/Resist		10			Silicon1 Top
6.3		Expose Material	Expose resist	/Organics/Resist		50	Gate	Light	Silicon1 Top
6.4		Enhanced Basic Etch	metal Etch	/Metals/Al /Organics/Resist		5			Silicon1 Top
6.5		Remove Materials	Remove resist	/Organics/Resist					Silicon1 Top
7		Sequence	Spacer pattern						
7.1		Conformal Deposit	Spacer deposition	/Nitrides/Si3N4 LPCVD		5			Silicon1 Top
7.2		Planarizing Deposit	resist for spacer pattern	/Organics/Resist		10			Silicon1 Top
7.3		Expose Material	Expose resist	/Organics/Resist		50	spacer	Light	Silicon1 Top
7.4		Enhanced Basic Etch	spacer Etch	/Nitrides/Si3N4 LPCVD /Organics/Resist		5			Silicon1 Top
7.5		Remove Materials	Remove resist	/Organics/Resist					Silicon1 Top
8		Enhanced Basic Etch	Spacer over Etch	/Nitrides/Si3N4 LPCVD		2			Silicon1 Top

Figure 2 : Process flow for spacer formation

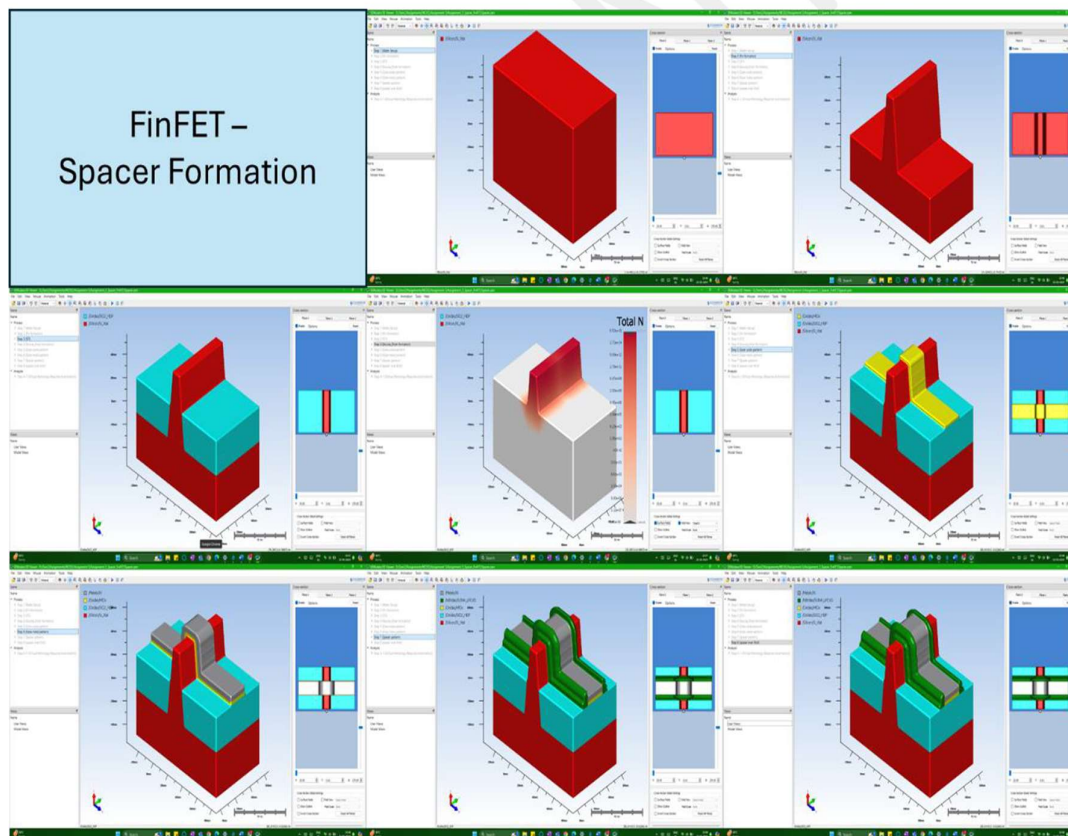


Figure 3 : Final structure for spacer formation

Question1(a)

The deposition technique used in the process is LPCVD for conformal deposition of spacers. It can be identified by material used is Si3N4_LPCVD as shown by Figure 4 below which means it is silicon nitride deposited using low pressure chemical vapour deposition (LPCVD) as can be seen in deposited structure Figure 5.

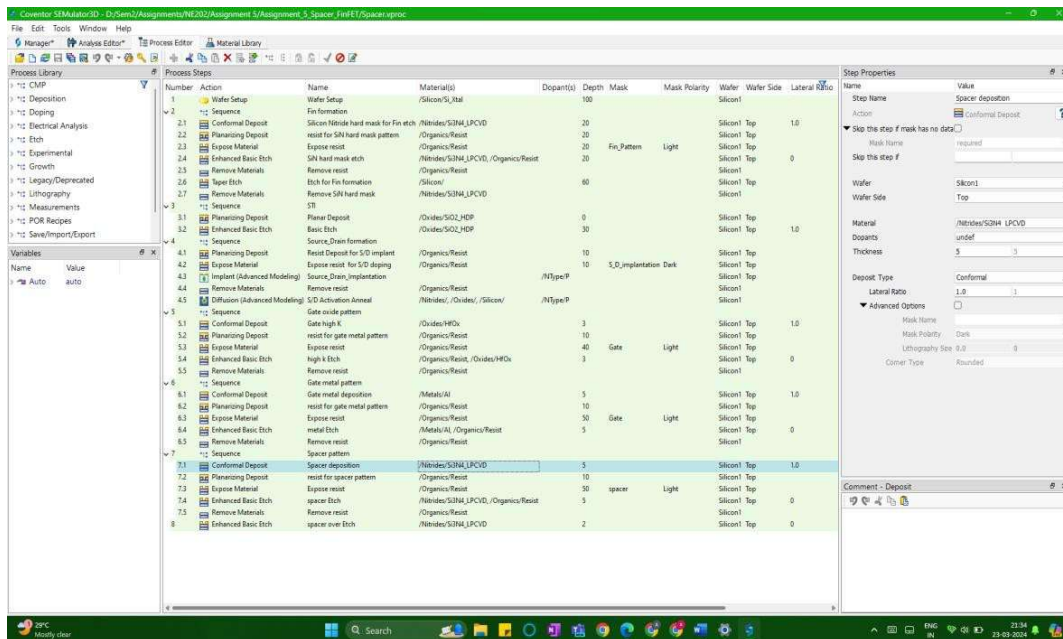


Figure 4 : Highlighting step for spacer deposition

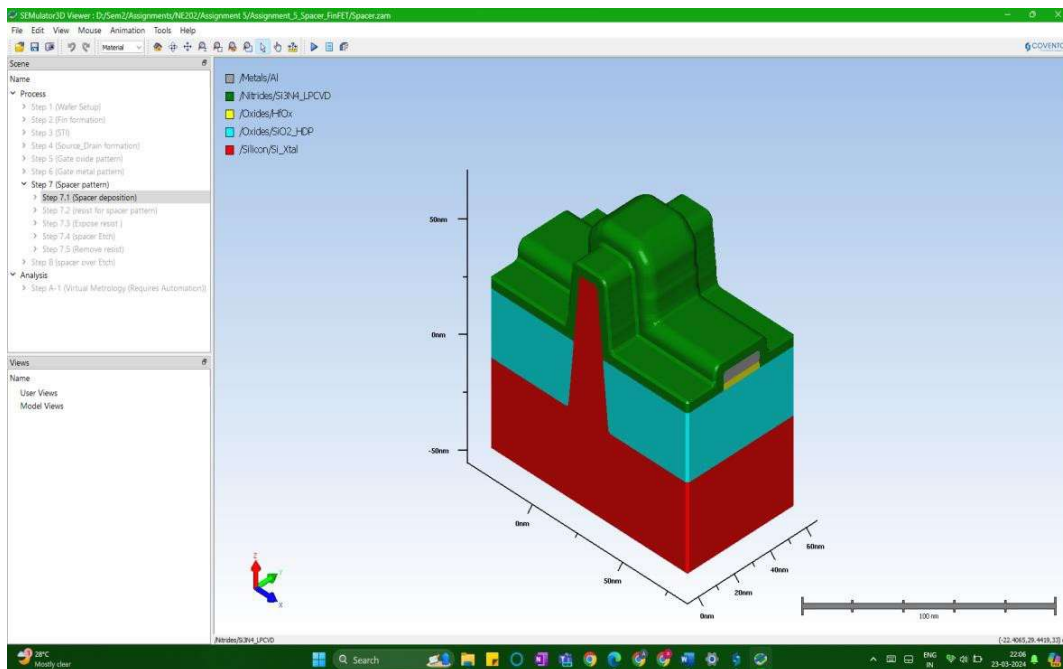


Figure 5 : Structure after Spacer deposition

Conformality is critically important for spacer deposition because of following,

- A conformally deposited spacer guarantees a uniform and well-defined channel width throughout the device.
- If the material doesn't fully coat the sidewalls of the fins, can lead to gaps or inconsistencies. These gaps can create leakage paths between the gate and the source/drain, allowing unwanted current flow and degrading device performance.
- Conformality ensures consistent spacer thickness even on high-aspect-ratio features like the thin sidewalls of fins.

Few conformal deposition techniques -

- Low-Pressure Chemical Vapor Deposition (LPCVD).
- Atomic Layer Deposition (ALD).
- Plasma-Enhanced Chemical Vapor Deposition (PECVD).
- Inductively coupled plasma chemical vapor deposition (ICPCVD).
- Silicide Spacer Formation.

Question1(b)

Critical Dimension refers to the minimum thickness or width that the spacer needs to achieve optimal device performance and functionality.

Using expeditor, we vary etch depth and observe spacer thickness, CD base and CD loss (by using, $CD_{loss} = CD_{base} - CD_{inflection_point}$), as seen in Figure 6.

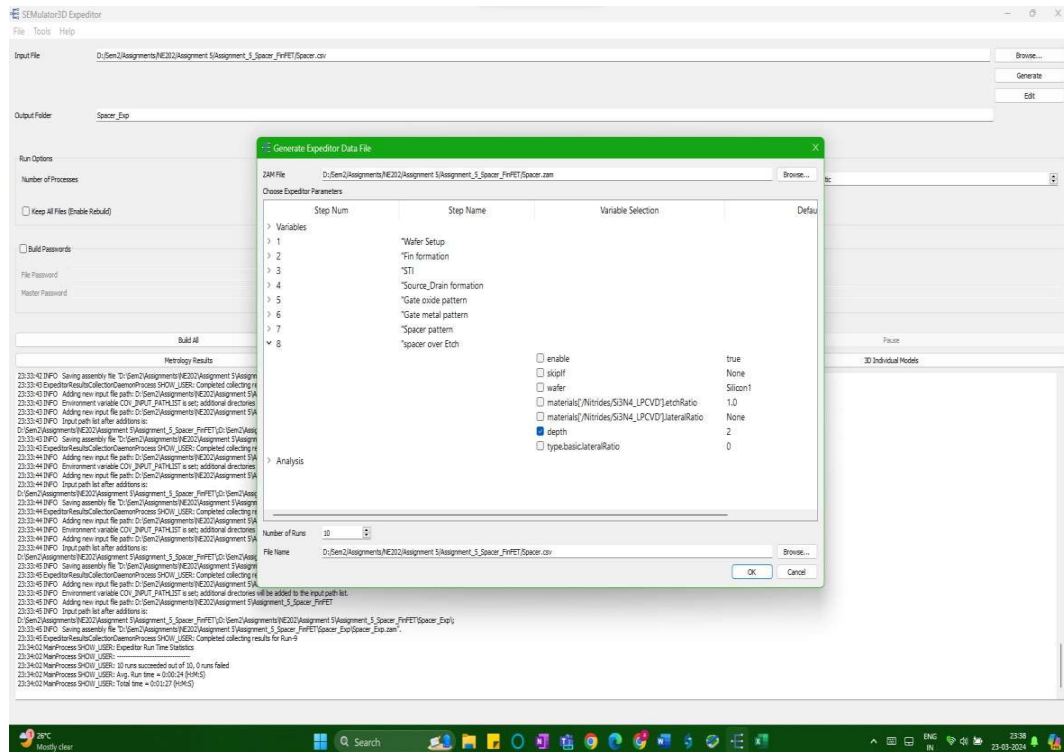


Figure 6 : Varying Etch depth using expeditor

CD_{base} , it is the first few nanometers of deposited spacer after etching. Here we are measuring 1 nm away from bottom of the surface to avoid and errors.

$CD_{inflection_point}$ is defined as critical dimension at which topography is seen to be curved. Here we have considered 50% high from bottom to top.

Lastly spacer thickness is measured by using film thickness measurement technique.

All the measurement parameters are shown by Figure 7.

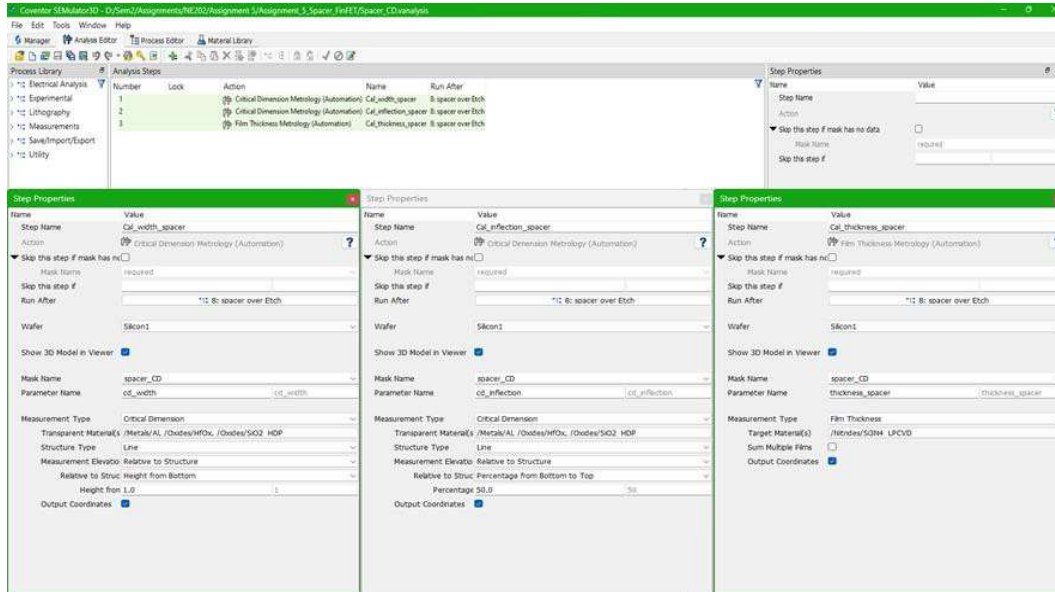


Figure 7 : Measurement parameters

Table 1 shows the various etch depths and all the parameters needed to be observed.

Table 1 : Variation in parameters wrt etch depth

Run No.	Etch depth (nm)	CD width (nm)	CD at inflection (nm)	CD loss (nm)	Thickness of spacer (nm)
Run-0	0	7.00	5.00	2.00	11.448
Run-1	1	7.00	5.00	2.00	10.386
Run-2	2	7.00	5.00	2.00	9.386
Run-3	3	6.98	5.00	1.98	8.386
Run-4	4	5.78	5.00	0.78	7.386
Run-5	5	4.58	5.00	-0.42	6.386
Run-6	5.5	4.48	4.87	-0.39	6.000
Run-7	6	4.38	4.81	-0.43	5.386
Run-8	6.5	4.28	4.71	-0.43	5.000
Run-9	7	4.18	4.59	-0.41	4.386

From table above, CD at inflection point is almost constant to 5 nm but CD at base decreases as we increase etch depth resulting in negative CD loss, which is not practically possible making **Run-5 to Run-9 invalid**.

Figure 8 shows the plotted results in Table 1. It is visible that if we increase etch depth beyond 4.5 nm then CD at inflection point is more than the CD at base which is not possible.

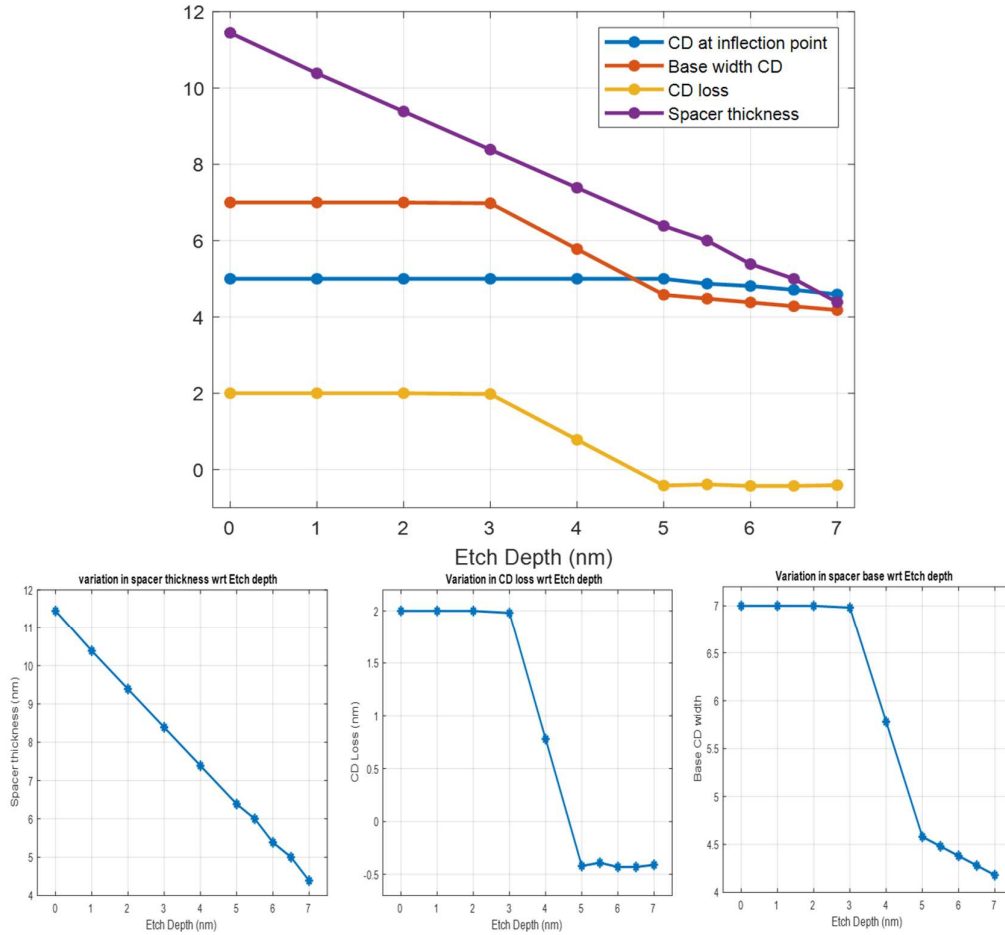


Figure 8 : Variation of spacer parameters wrt Etch Depth

Increasing over-etch reduces the thickness of the spacer. The etching process removes material not just from the target area but also from the spacer.

Initially, increasing over-etch have minimal impact on the CD_base as well as CD_loss. This is because the etch process might prioritize removing material from the exposed areas (outside the spacer) before significantly affecting the underlying feature (protected by the spacer). **Later, with further increase in over-etch, leads to a decrease in both CD_base and CD_loss.** The larger etch depth starts to remove material from the desired feature itself, reducing its size.

Question1(c)

If CD = 7 nm with over etch depth = 2 nm then CD_loss can be calculated by,

$$CD_{loss} = CD_{base} - CD_{inflection_point}$$

Figure 9 and Figure 10 shows the metrology results for Critical dimension measurement at different points.

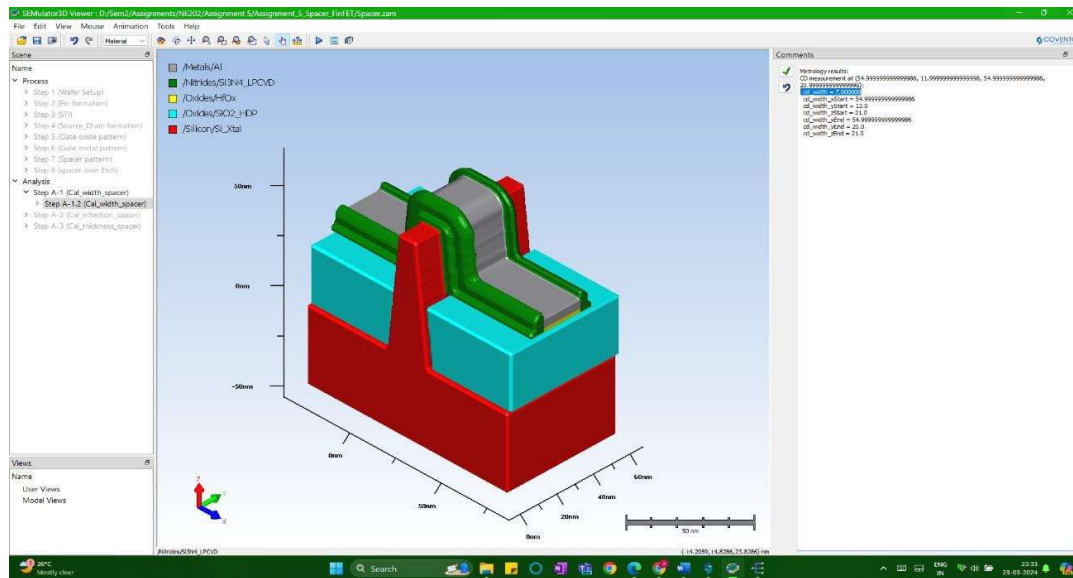


Figure 9 : Spacer base critical dimension measurement

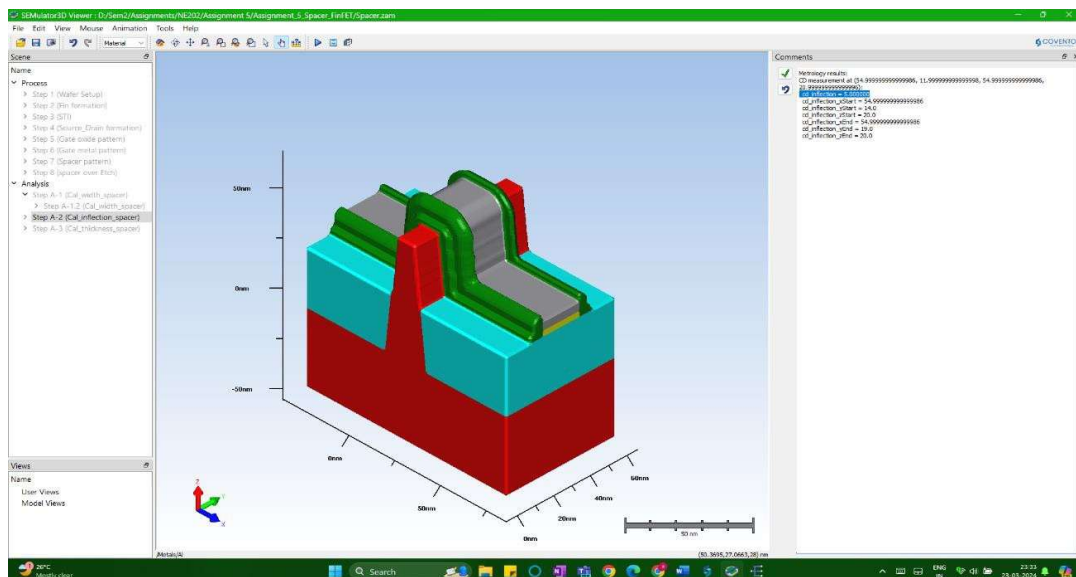


Figure 10 : Critical dimension measurement at inflection point

$$CD_{loss} = 7 - 5 = 2nm$$

Here, measured thickness of spacer is 9.386 nm.

Question1(d)

Figure 11 shows the CD profile in simulated structure and signifies the shape of the spacer along the gate height.

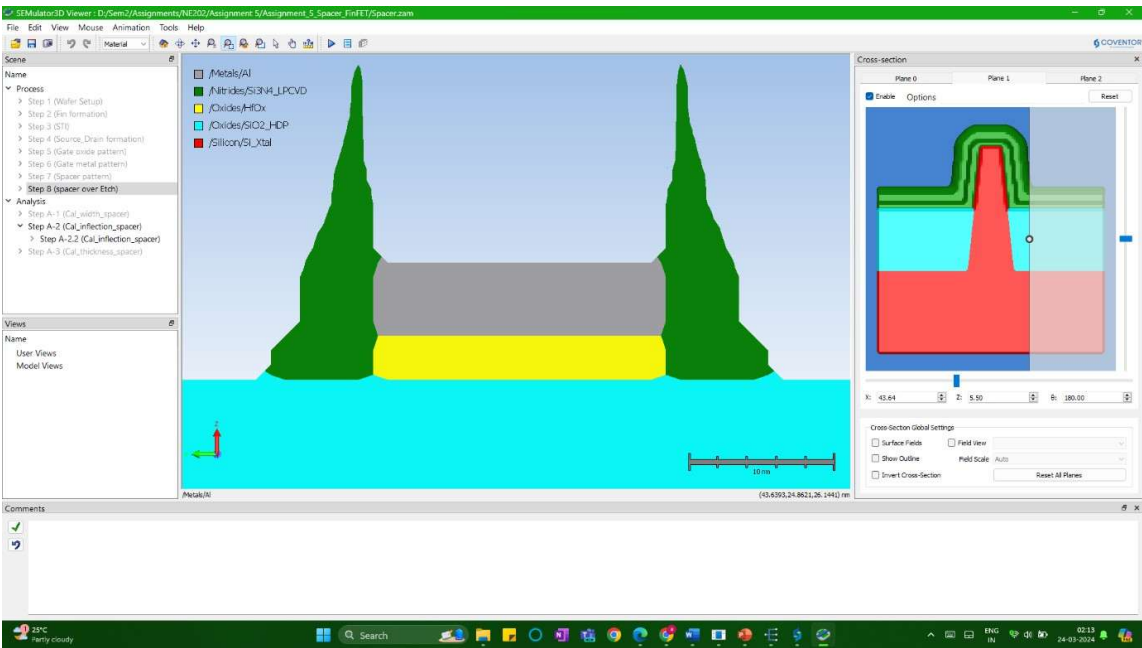


Figure 11 : Simulated CD profile

Using expeditor (Figure 12) we measure CD at different points on along the gate to get our profile.

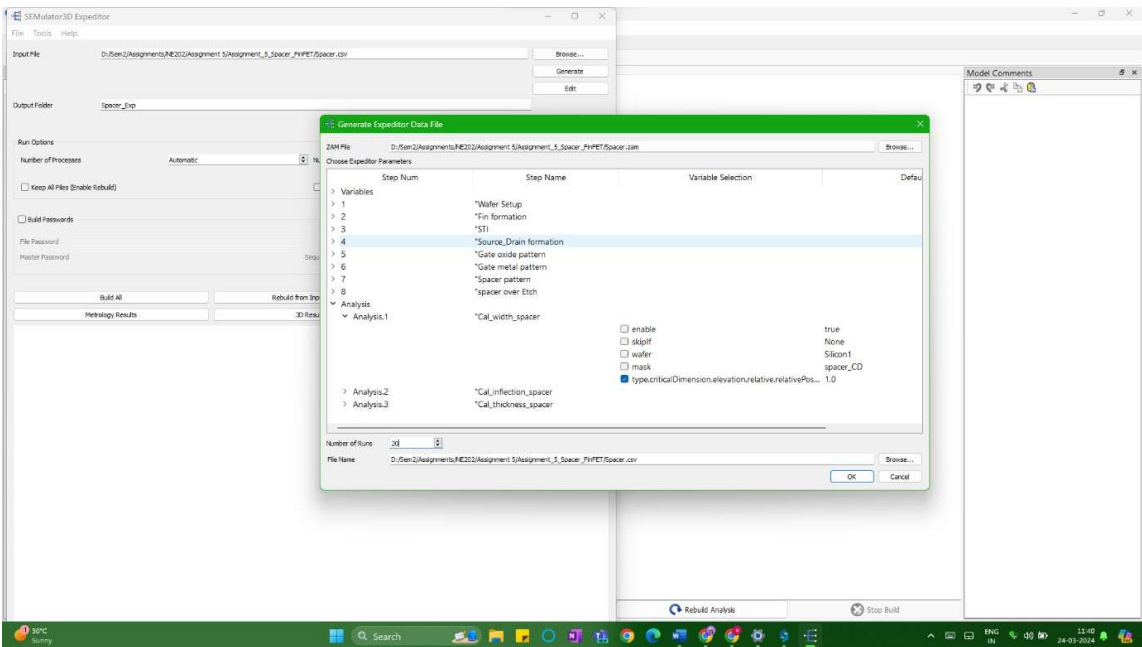


Figure 12 : Varying measurement distance using expeditor

The results obtained from the expeditor run is shown by Table 2 below,

Table 2 : Variation in CD along gate height

Run	Gate thickness (nm)	CD width (nm)
Run-0	0.0	5.00
Run-1	0.5	6.00
Run-2	1.0	7.00
Run-3	1.5	7.00
Run-4	2.0	7.00
Run-5	2.5	6.50
Run-6	3.0	6.00
Run-7	3.5	5.50
Run-8	4.0	5.00
Run-9	4.5	5.00
Run-10	5.0	5.00
Run-11	5.5	5.00
Run-12	6.0	5.00
Run-13	6.5	5.00
Run-14	7.0	5.00
Run-15	7.5	4.78
Run-16	8.0	4.57
Run-17	8.5	4.21
Run-18	9.0	3.84
Run-19	9.5	3.40

Plotting the results obtained in Table 2 gives us curve as shown in Figure 13.

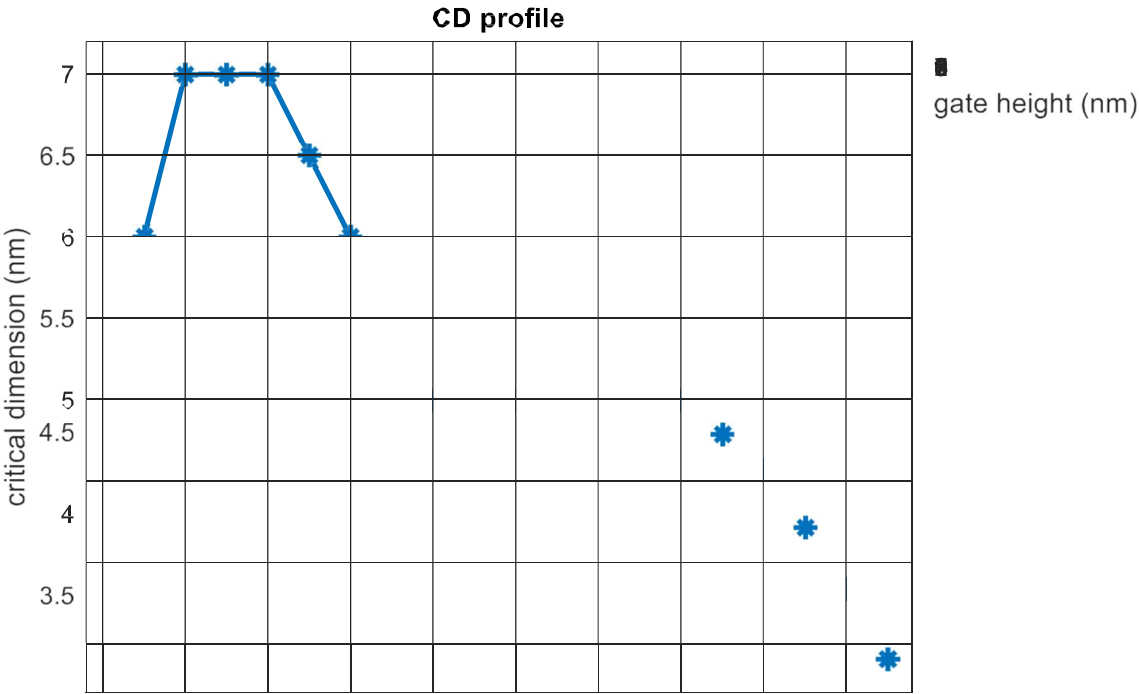


Figure 13 : CD profile

On comparing both the results (Figure 14) we see the shape is same for both.

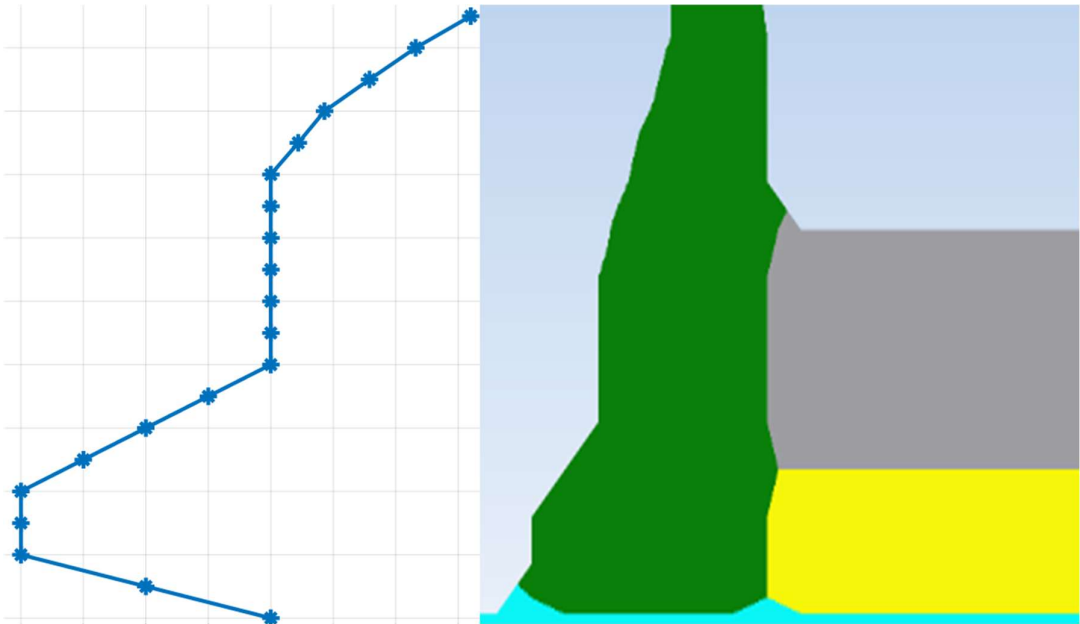


Figure 14 : Comparison of simulated and measured CD profile

Question 2: 14nm advanced logic process

This task is centred around 14 nm advanced logic process, we aim to pinpoint the specific steps where spacer deposition and patterning occur. Once the spacer formation steps are identified, we will explore the materials utilized for spacer formation. This analysis will provide insights into the integration of spacers within the overall FET fabrication sequence.

Figure 15 shows process flow for this simulation.

Step	Name	Material	Description	Mask	Mask Priority	Mode	Wafer Size	Lot Size
1	Pre-Deposition	SiO2	Pre-Deposition	None	None	None	300mm	100
2	Deposition	SiO2	Deposition	None	None	None	300mm	100
3	Etch	SiO2	Etch	None	None	None	300mm	100
4	Growth	SiO2	Growth	None	None	None	300mm	100
5	Lithography	SiO2	Lithography	None	None	None	300mm	100
6	Measurement	SiO2	Measurement	None	None	None	300mm	100
7	WIP Reagents	SiO2	WIP Reagents	None	None	None	300mm	100
8	Transfer/Transfer/Transfer	SiO2	Transfer/Transfer/Transfer	None	None	None	300mm	100
9	Pre-Deposition	SiO2	Pre-Deposition	None	None	None	300mm	100
10	Deposition	SiO2	Deposition	None	None	None	300mm	100
11	Etch	SiO2	Etch	None	None	None	300mm	100
12	Growth	SiO2	Growth	None	None	None	300mm	100
13	Lithography	SiO2	Lithography	None	None	None	300mm	100
14	Measurement	SiO2	Measurement	None	None	None	300mm	100
15	WIP Reagents	SiO2	WIP Reagents	None	None	None	300mm	100
16	Transfer/Transfer/Transfer	SiO2	Transfer/Transfer/Transfer	None	None	None	300mm	100
17	Pre-Deposition	SiO2	Pre-Deposition	None	None	None	300mm	100
18	Deposition	SiO2	Deposition	None	None	None	300mm	100
19	Etch	SiO2	Etch	None	None	None	300mm	100
20	Growth	SiO2	Growth	None	None	None	300mm	100
21	Lithography	SiO2	Lithography	None	None	None	300mm	100
22	Measurement	SiO2	Measurement	None	None	None	300mm	100
23	WIP Reagents	SiO2	WIP Reagents	None	None	None	300mm	100
24	Transfer/Transfer/Transfer	SiO2	Transfer/Transfer/Transfer	None	None	None	300mm	100
25	Pre-Deposition	SiO2	Pre-Deposition	None	None	None	300mm	100
26	Deposition	SiO2	Deposition	None	None	None	300mm	100
27	Etch	SiO2	Etch	None	None	None	300mm	100
28	Growth	SiO2	Growth	None	None	None	300mm	100
29	Lithography	SiO2	Lithography	None	None	None	300mm	100
30	Measurement	SiO2	Measurement	None	None	None	300mm	100
31	WIP Reagents	SiO2	WIP Reagents	None	None	None	300mm	100
32	Transfer/Transfer/Transfer	SiO2	Transfer/Transfer/Transfer	None	None	None	300mm	100
33	Pre-Deposition	SiO2	Pre-Deposition	None	None	None	300mm	100
34	Deposition	SiO2	Deposition	None	None	None	300mm	100
35	Etch	SiO2	Etch	None	None	None	300mm	100
36	Growth	SiO2	Growth	None	None	None	300mm	100
37	Lithography	SiO2	Lithography	None	None	None	300mm	100
38	Measurement	SiO2	Measurement	None	None	None	300mm	100
39	WIP Reagents	SiO2	WIP Reagents	None	None	None	300mm	100
40	Transfer/Transfer/Transfer	SiO2	Transfer/Transfer/Transfer	None	None	None	300mm	100
41	Pre-Deposition	SiO2	Pre-Deposition	None	None	None	300mm	100
42	Deposition	SiO2	Deposition	None	None	None	300mm	100
43	Etch	SiO2	Etch	None	None	None	300mm	100
44	Growth	SiO2	Growth	None	None	None	300mm	100
45	Lithography	SiO2	Lithography	None	None	None	300mm	100
46	Measurement	SiO2	Measurement	None	None	None	300mm	100
47	WIP Reagents	SiO2	WIP Reagents	None	None	None	300mm	100
48	Transfer/Transfer/Transfer	SiO2	Transfer/Transfer/Transfer	None	None	None	300mm	100
49	Pre-Deposition	SiO2	Pre-Deposition	None	None	None	300mm	100
50	Deposition	SiO2	Deposition	None	None	None	300mm	100
51	Etch	SiO2	Etch	None	None	None	300mm	100
52	Growth	SiO2	Growth	None	None	None	300mm	100
53	Lithography	SiO2	Lithography	None	None	None	300mm	100
54	Measurement	SiO2	Measurement	None	None	None	300mm	100
55	WIP Reagents	SiO2	WIP Reagents	None	None	None	300mm	100
56	Transfer/Transfer/Transfer	SiO2	Transfer/Transfer/Transfer	None	None	None	300mm	100
57	Pre-Deposition	SiO2	Pre-Deposition	None	None	None	300mm	100
58	Deposition	SiO2	Deposition	None	None	None	300mm	100
59	Etch	SiO2	Etch	None	None	None	300mm	100
60	Growth	SiO2	Growth	None	None	None	300mm	100
61	Lithography	SiO2	Lithography	None	None	None	300mm	100
62	Measurement	SiO2	Measurement	None	None	None	300mm	100
63	WIP Reagents	SiO2	WIP Reagents	None	None	None	300mm	100
64	Transfer/Transfer/Transfer	SiO2	Transfer/Transfer/Transfer	None	None	None	300mm	100
65	Pre-Deposition	SiO2	Pre-Deposition	None	None	None	300mm	100
66	Deposition	SiO2	Deposition	None	None	None	300mm	100
67	Etch	SiO2	Etch	None	None	None	300mm	100
68	Growth	SiO2	Growth	None	None	None	300mm	100
69	Lithography	SiO2	Lithography	None	None	None	300mm	100
70	Measurement	SiO2	Measurement	None	None	None	300mm	100
71	WIP Reagents	SiO2	WIP Reagents	None	None	None	300mm	100
72	Transfer/Transfer/Transfer	SiO2	Transfer/Transfer/Transfer	None	None	None	300mm	100
73	Pre-Deposition	SiO2	Pre-Deposition	None	None	None	300mm	100
74	Deposition	SiO2	Deposition	None	None	None	300mm	100
75	Etch	SiO2	Etch	None	None	None	300mm	100
76	Growth	SiO2	Growth	None	None	None	300mm	100
77	Lithography	SiO2	Lithography	None	None	None	300mm	100
78	Measurement	SiO2	Measurement	None	None	None	300mm	100
79	WIP Reagents	SiO2	WIP Reagents	None	None	None	300mm	100
80	Transfer/Transfer/Transfer	SiO2	Transfer/Transfer/Transfer	None	None	None	300mm	100
81	Pre-Deposition	SiO2	Pre-Deposition	None	None	None	300mm	100
82	Deposition	SiO2	Deposition	None	None	None	300mm	100
83	Etch	SiO2	Etch	None	None	None	300mm	100
84	Growth	SiO2	Growth	None	None	None	300mm	100
85	Lithography	SiO2	Lithography	None	None	None	300mm	100
86	Measurement	SiO2	Measurement	None	None	None	300mm	100
87	WIP Reagents	SiO2	WIP Reagents	None	None	None	300mm	100
88	Transfer/Transfer/Transfer	SiO2	Transfer/Transfer/Transfer	None	None	None	300mm	100
89	Pre-Deposition	SiO2	Pre-Deposition	None	None	None	300mm	100
90	Deposition	SiO2	Deposition	None	None	None	300mm	100
91	Etch	SiO2	Etch	None	None	None	300mm	100
92	Growth	SiO2	Growth	None	None	None	300mm	100
93	Lithography	SiO2	Lithography	None	None	None	300mm	100
94	Measurement	SiO2	Measurement	None	None	None	300mm	100
95	WIP Reagents	SiO2	WIP Reagents	None	None	None	300mm	100
96	Transfer/Transfer/Transfer	SiO2	Transfer/Transfer/Transfer	None	None	None	300mm	100
97	Pre-Deposition	SiO2	Pre-Deposition	None	None	None	300mm	100
98	Deposition	SiO2	Deposition	None	None	None	300mm	100
99	Etch	SiO2	Etch	None	None	None	300mm	100
100	Growth	SiO2	Growth	None	None	None	300mm	100

Figure 15 : Steps involved in FinFET simulation

Resultant structure is shown by Figure 16 and Figure 17.

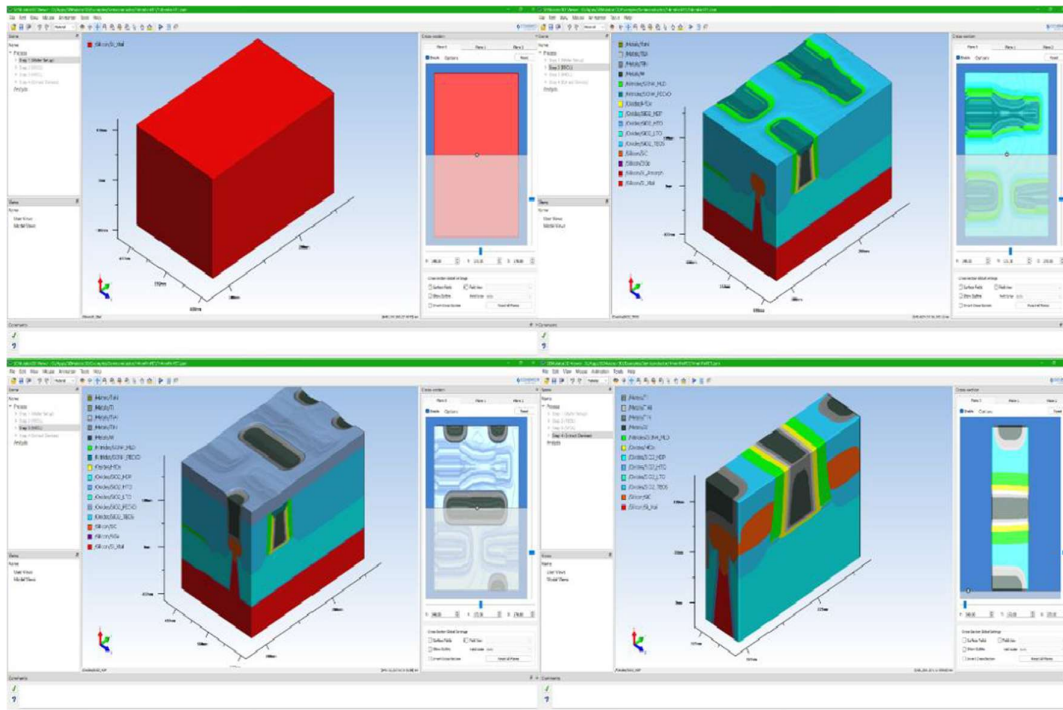


Figure 16 : Simulated 14nm FinFET process (3D view)

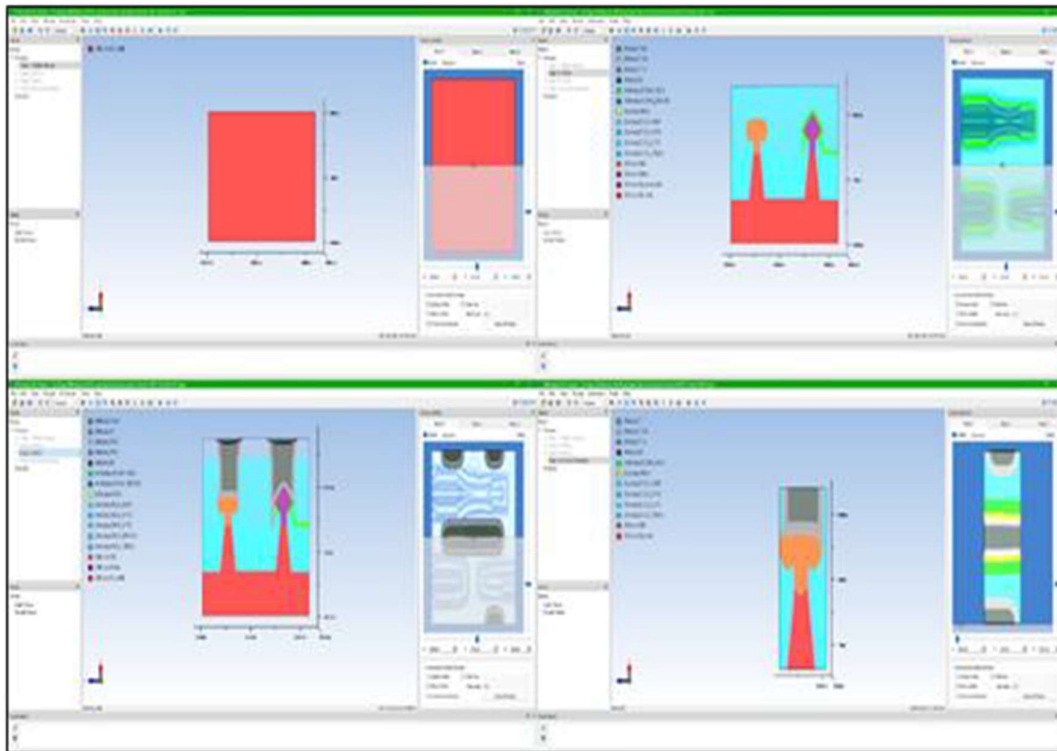


Figure 17 : Simulated 14nm FinFET process (Front View)

Question2(a)

Spacers are formed in **STEP 2.6** for both NMOS and PMOS while before implanting source and drain. The sub steps are as follows,

For PMOS (Figure 18 and Figure 19),

- Step 2.6.1.1 – Deposition of LTO oxide.
- Step 2.6.1.2 – Deposition of MLD nitride.
- Step 2.6.1.3 – Photoresist Coating.
- Step 2.6.1.4 – Photoresist pattern.
- Step 2.6.1.5 – Shaping spacers.

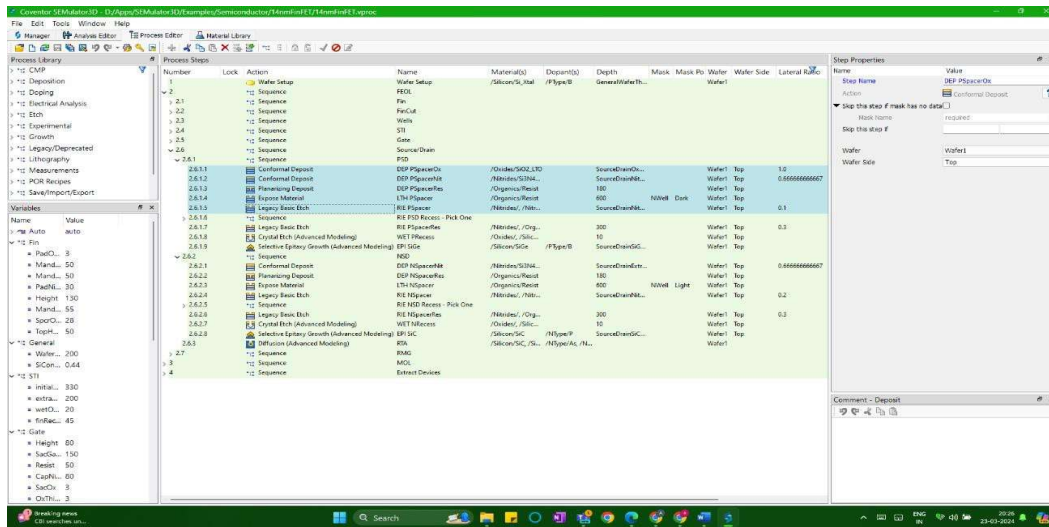


Figure 18 : Steps for Spacer formation for PMOS

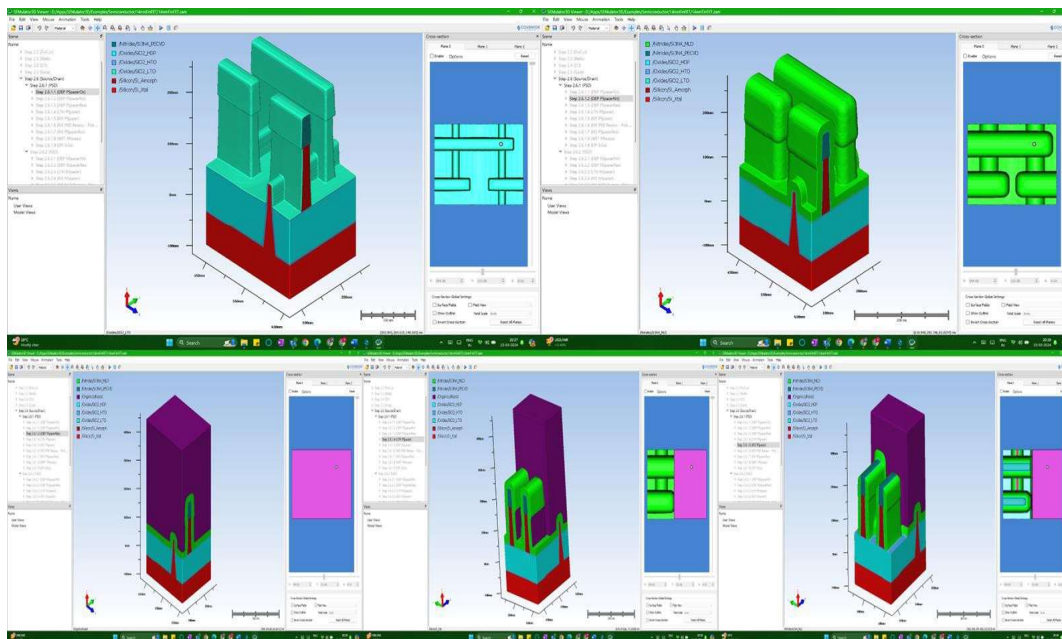


Figure 19 : Simulated results for Spacer formation for PMOS

For NMOS (Figure 20 and Figure 21),

- Step 2.6.2.1 – Deposition of MLD nitride.
- Step 2.6.2.2 – Photoresist coating.
- Step 2.6.2.3 – Photoresist pattern.
- Step 2.6.2.4 – Shaping spacers.

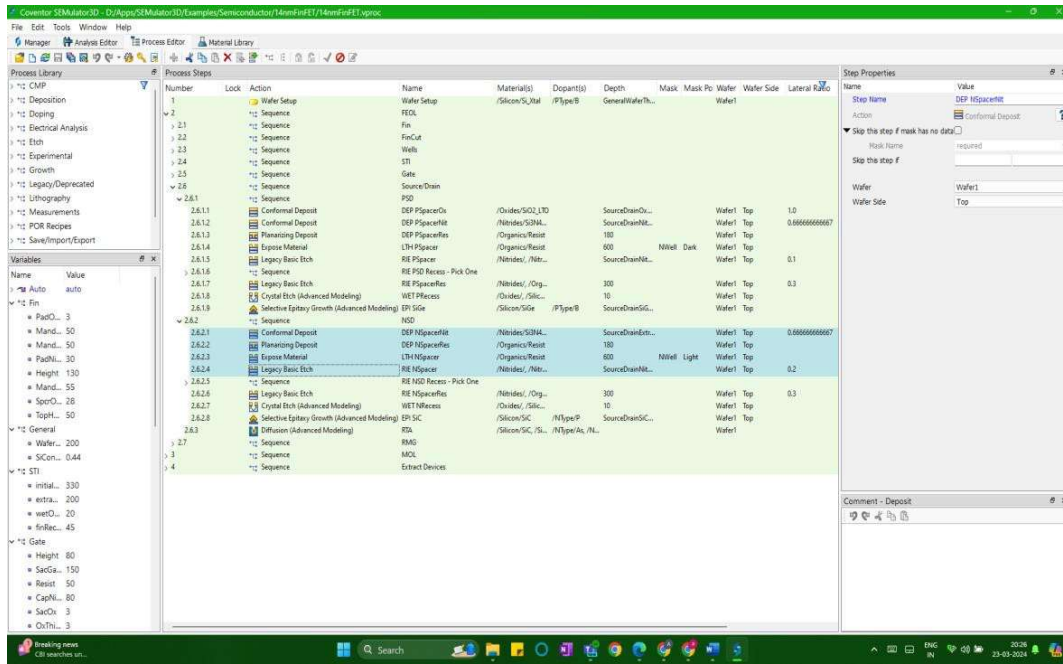


Figure 20 : Steps for Spacer formation for NMOS

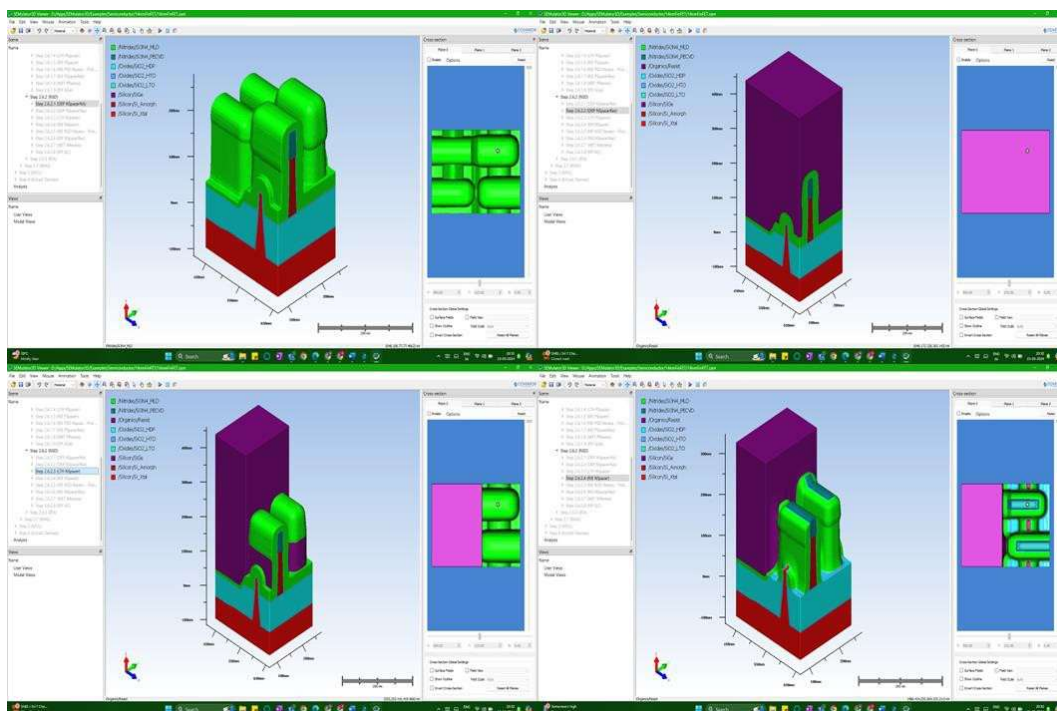


Figure 21: Simulated results for Spacer formation for NMOS

Question2(b)

To minimize electrical interference between the gate and source/drain areas of a transistor, spacer materials with low electrical resistance (low dielectric constant) are preferred. This reduces the unwanted capacitance between these elements. The material used should have conformality, high temperature tolerance, selectivity during etching.

Here, First a layer of SiO₂ is deposited and after that SiN is deposited one by one on each MOS.

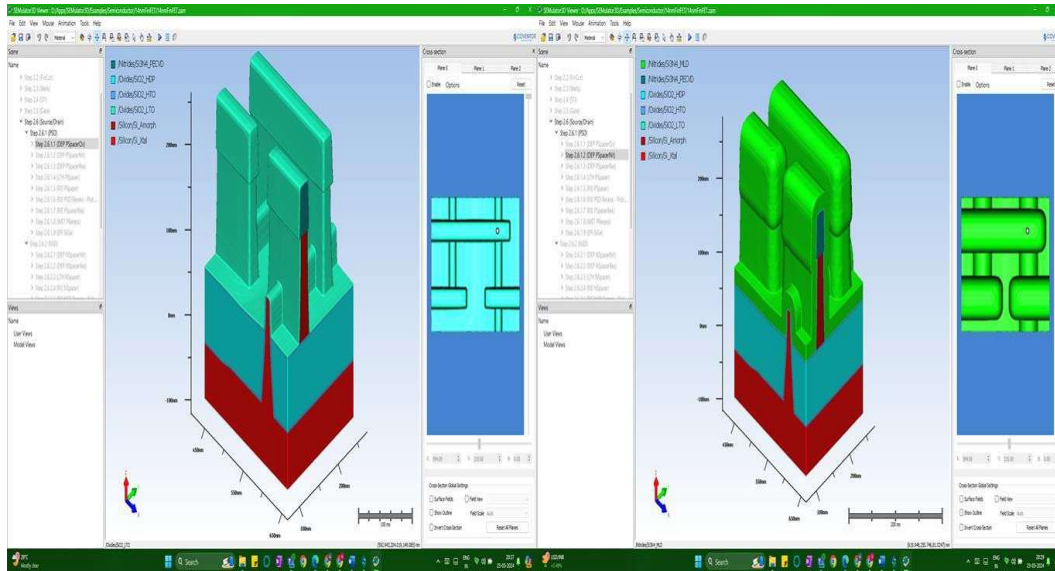


Figure 22 : Material used for spacer in PMOS

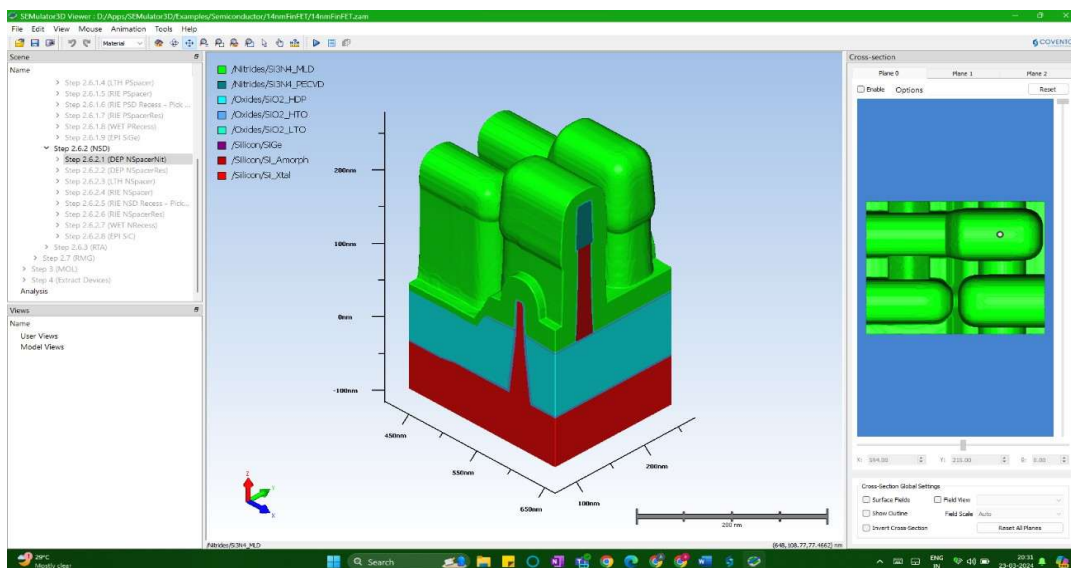


Figure 23 : Material used for spacer in NMOS

Conclusion

Spacers are essential components in FET fabrication, serving multiple purposes. They define the channel region, crucial for controlling current flow, and provide electrical isolation between the source and drain electrodes. As FETs continue to miniaturize, spacer design becomes increasingly important. The orientation of fins and spacers further influences device characteristics. Understanding these aspects is vital for optimizing FET performance and pushing the boundaries of transistor scaling.

References

- [1] <https://www.numerade.com/ask/question/1-name-the-mosfet-in-the-diagram-below-what-is-the-type-of-the-mosfet-in-the-diagram-below-4marks-2-explain-with-diagram-each-semiconductor-fabrication-process-in-order-to-produce-the-mosfet-96817/>
- [2] <https://iopscience.iop.org/article/10.1088/0268-1242/27/4/045004/pdf>
- [3] Kang, S. W. (Ed.). (2006). Handbook of thin film deposition processes and techniques (2nd ed.). Noyes Publications/William Andrew Publishing.
- [4] <https://www.sciencedirect.com/science/article/abs/pii/S1385894722001760>
- [5] Gu, M., Wang, X., Li, W., Aquilino, M., Peng, J., Wang, H., Jaeger, D., Tabakman, K., Carter, R., Hu, O., Ma, W., Joshi, M., & Lee, L. (2020). Hybrid low- k spacer scheme for advanced FinFET technology parasitic capacitance reduction. Electronics Letters, 56(10), 514–516. <https://doi.org/10.1049/el.2019.3954>